

PROCESS FOLLOWED

Initially the dataset was downloaded from the official website which had around 33 categories and there are 63 individual datasets which are merged and each one had around 712K Rows and initially a drawback was seen which was when the data was loaded into google colab and an attempt has been made to merge all of them into a single file, then even colab crashed due to ram error and the step was to go through whole of the data and to get it separated by some samples and new samples were made using **stratified sampling** and was taken only 5% of the data which keeps the hold of all the types of attack and here we considered only 5% on second thought, while first it was kept down to 10% and 5% was chosen because each dataset had roughly 700K rows which goes down to 70K rows even when sampled by 10% and when this is done and merged for 63 files, it would be around 4.2M and this would be a very large dataset for the model to run on it and for the sake of project, to keep the dataset small, we used only 5% which keeps it down to 2.1M parameters which is still considered to be a lot in case of project and research and this can be used to train the model better and to test it in the later case and 5% of data was sampled from each dataset and was put into a single dataset which is a combined one.

And once the data has been merged then the thing that has been done is checking for the duplicates and then dropping them if there are any left in the dataset that has been merged and once if there are no duplicates left, then that particular dataset is saved as the latest processed dataset. Followed by that we will check if there are any missing values in any of the labels and will print them and in this type of datasets, there are chances that some of the fields might be filled with something bw +infinity to -infinity and then we will replace all the inf numbers that have been found with NaN and we removed the missing values too as both of them combined together make upto 87 rows out of 2.1M rows which is a negligible difference.

Once we are done till the above step, we start off with EDA which is Exploratory Data Analysis and for this we will check with the number of available rows, columns, and we will even describe the dataset to get the mean, count, std deviation and once we are done with it we will plot the label so as to see the visual representation of the attacks and the count of that particular data and then we will even plot it for binary classification and then check the number of unique features that are available. Once it has been checked, then we will plot the correlated features on the heatmap so as to see the correlation between them so as to help PCC model while choosing the features and then we will start plotting the features using a histogram and instead of plotting all of them at once, we start with plotting 5 important features at once so as to see how they are distributed and once it is done then we will use boxplot so as to get the outlier detection of the dataset.

These can be inserted in our report from the EDA part

Dataset Overview (rows, columns, features)
Statistical Summary (describe())
Attack Class Distribution (bar chart)
Binary Distribution (Benign vs Attack)
Correlation Heatmap
Feature Distribution Histograms
Boxplots for Key Features

And once EDA has been done, we will be going ahead with Feature selection for which we will be doing it using PCC as specified in our literature survey

FEATURE SELECTION USING PCC

We will load the dataset or our csv file again using pandas and we will drop the label and we will check what is the shape of the dataframe and once that is done, then we will check the correlation matrix of that particular dataframe so as to see the number of rows and columns that are present in the dataframe which is cleaned and the features are to be selected and we will keep only the upper triangular matrix because the

features will be repeated and we don't need to have the whole data frame which will be too big for PCC and a np function **triu** is used which is used to get the upper triangular matrix of the correlation matrix and another matrix of same size as Correlation matrix has been created with all 1's in it apart from the diagonal and then it is filled with Boolean values in the matrix and then we will run a loop to compare if there are any features that exceed the threshold of 0.90 and we drop that features that are almost closely related and once done, we will get the count of the features that are to be dropped and we will drop them and get the correlation matrix and we will cross check the matrix again to see that the number of features that are available have been reduced

FEATURES TO BE REMOVED AFTER PCC ARE

Feature	Reason
rst_flag_number	Redundant with rst_count.
ack_count	Highly correlated with ACK-related packet statistics.
syn_count	Redundant with SYN flag information.
fin_count	Redundant with FIN flag information.
rst_count	Redundant after removing correlated RST flag features.
TCP	Highly correlated with Protocol Type. The protocol number already encodes TCP/UDP/ICMP.
IPv	Usually constant or strongly correlated with protocol/network fields.
LLC	Low additional information and highly correlated.
Std	Highly correlated with Variance (Variance = Std ²). Keeping both is unnecessary.
Tot size	Correlated with Tot sum; keeping only one avoids redundancy.

Once we have removed the features, then we need to scale the data so as to keep the values between 0 and 1 so as to make it easy for the data to be evaluated and for this we are using min max scaler to keep all the values accessible and once it is done we again checked by printing the ranges

Once that is done then we need to encode the data and we need to create separate datasets for the attack types and for this purpose we created a binary dataset which had high class imbalance where the original attack was around 96.44% and the remaining 3.56% was of benign and to overcome this we will be using some techniques only with the training dataset after test train split.